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(54)
SERVER-BASED METHOD OF USING A TRAINED DEEP LEARNING MODEL AND GROUND TRUTH LABELS

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None
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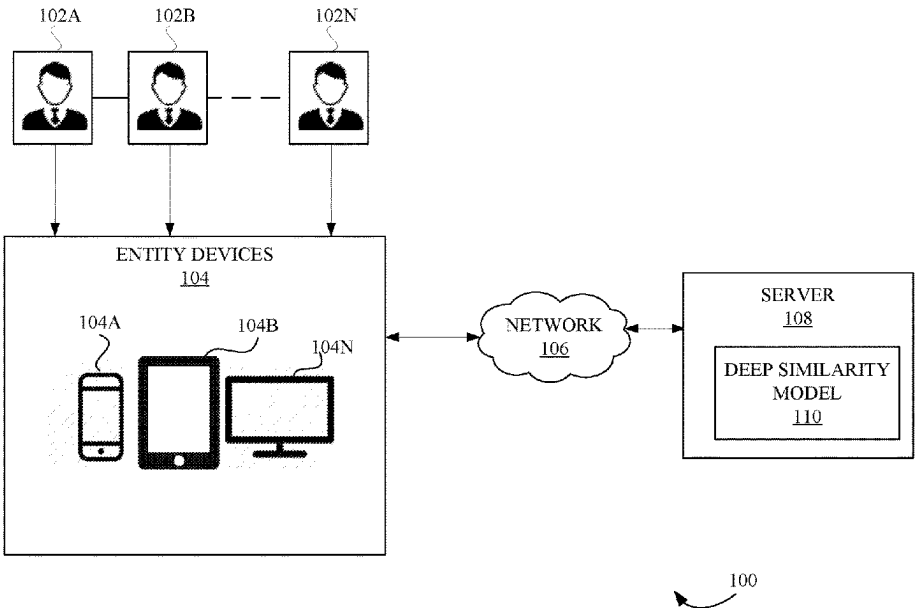
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(57)
ABSTRACT

A server-implemented method for determining, at a server, using a deep learning model, a cluster of device identifiers associated with computing devices having characteristics that are related to a match set of computing devices based on locations data streams and ground truth labels.

17 Claims, 13 Drawing Sheets



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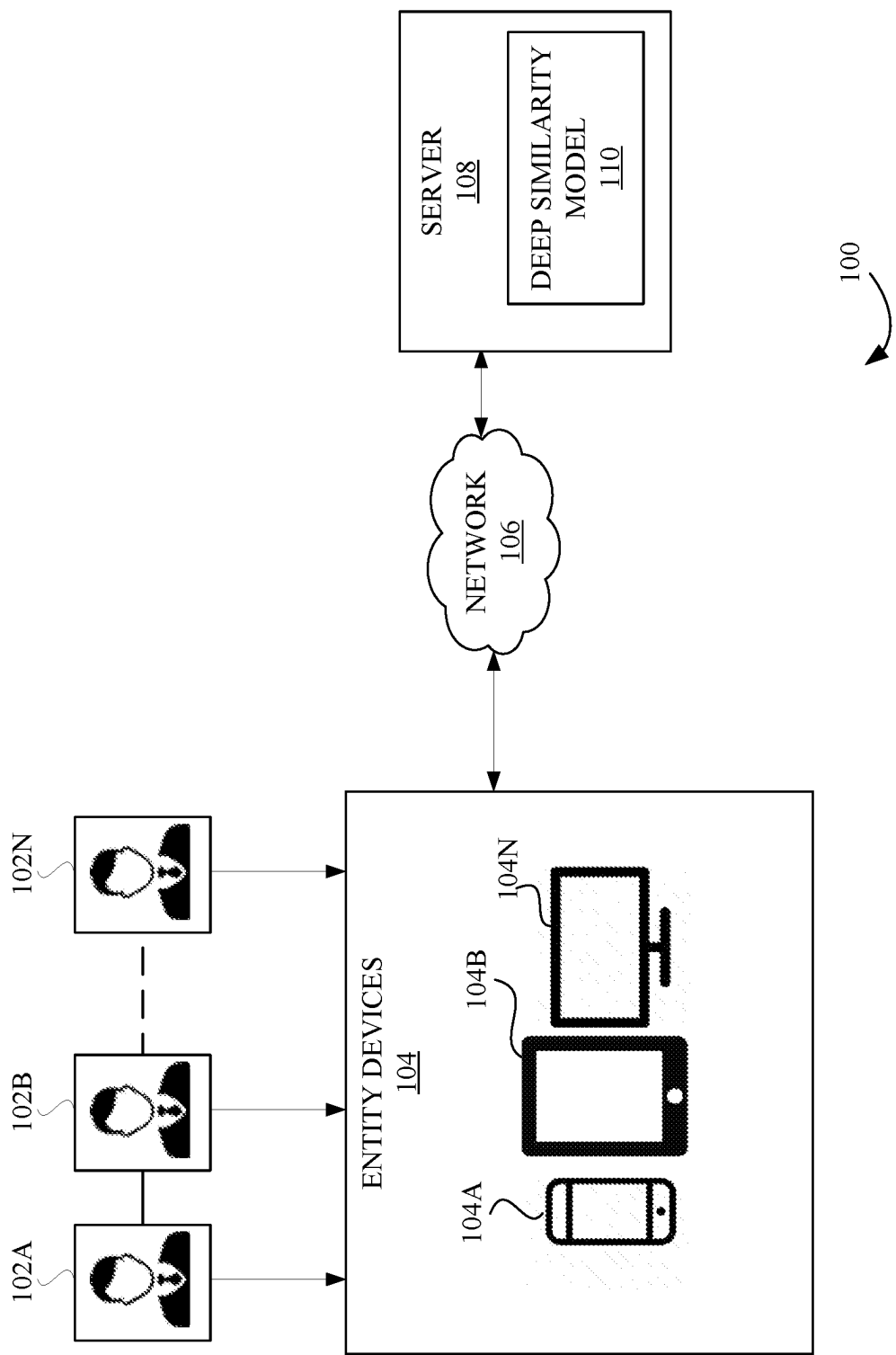
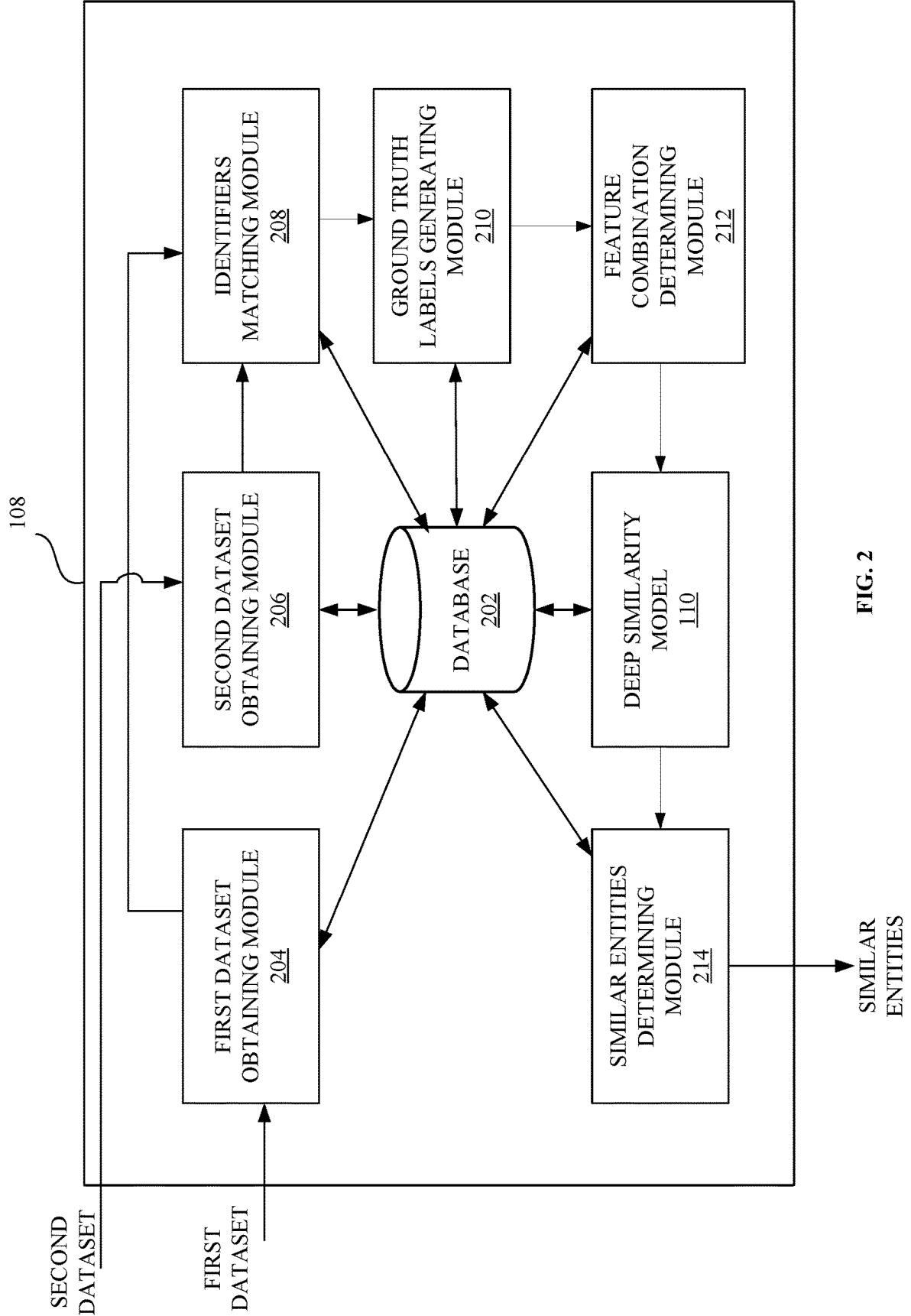


FIG. 1



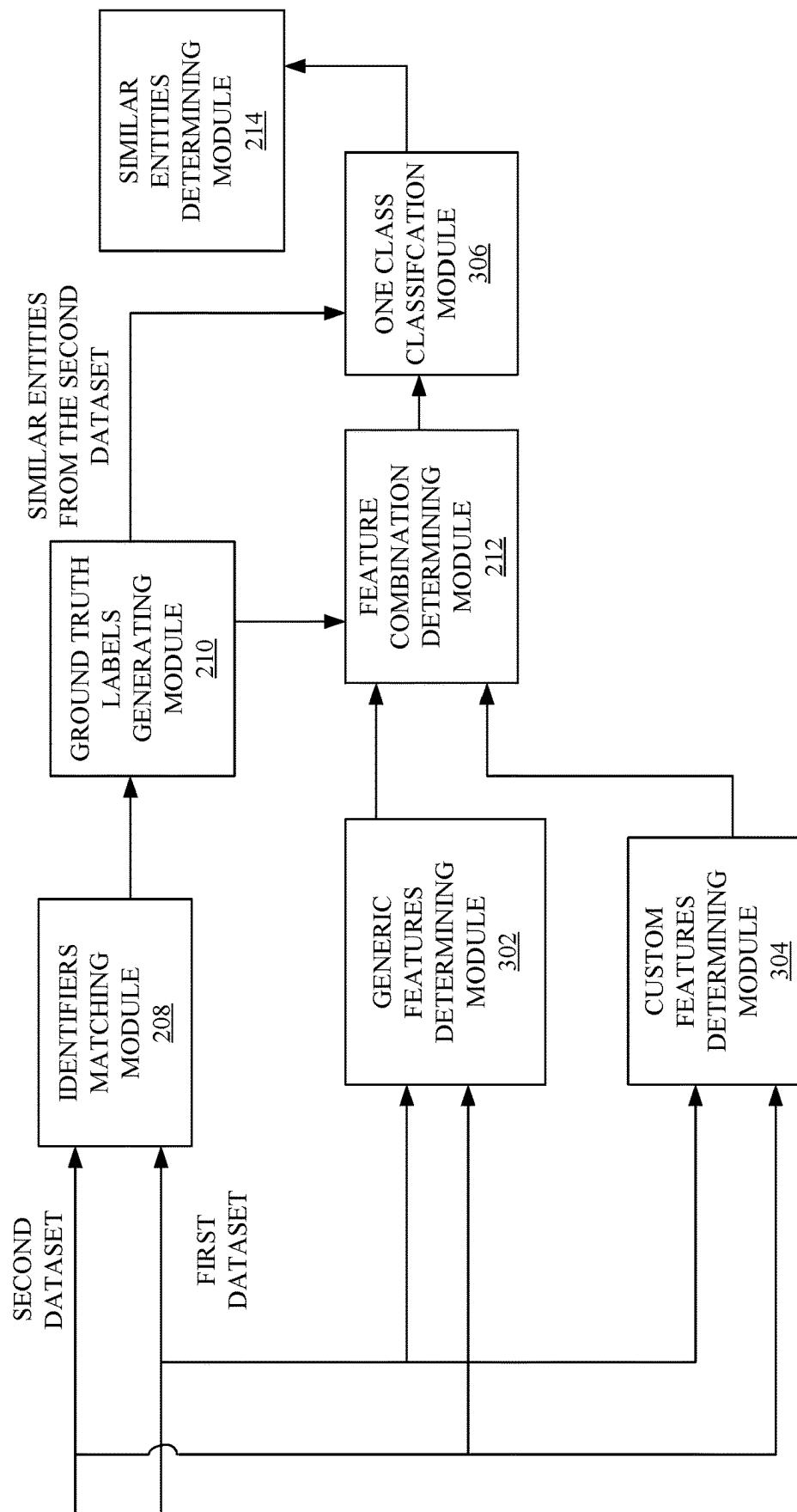


FIG. 3A

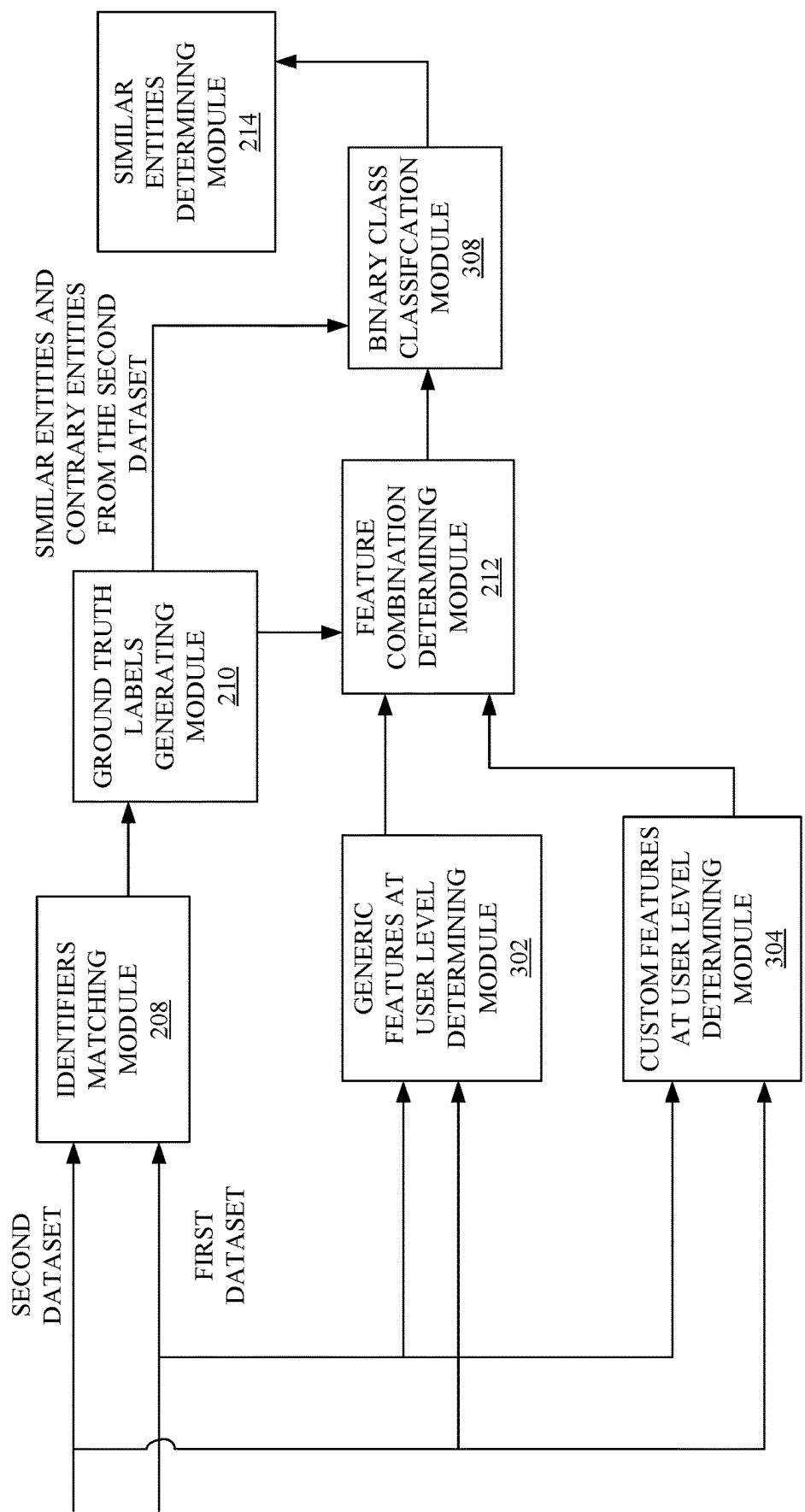


FIG. 3B

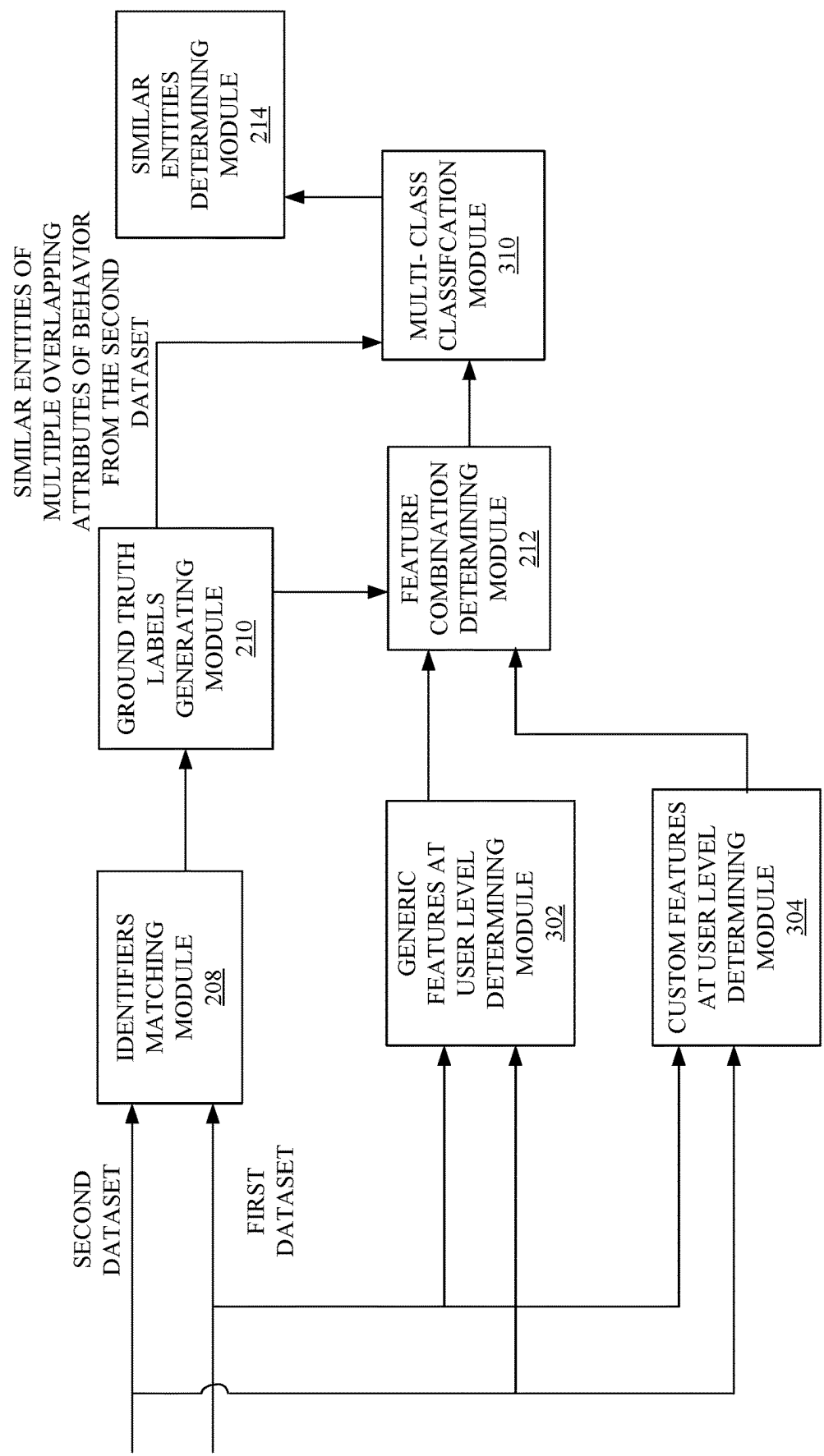


FIG. 3C

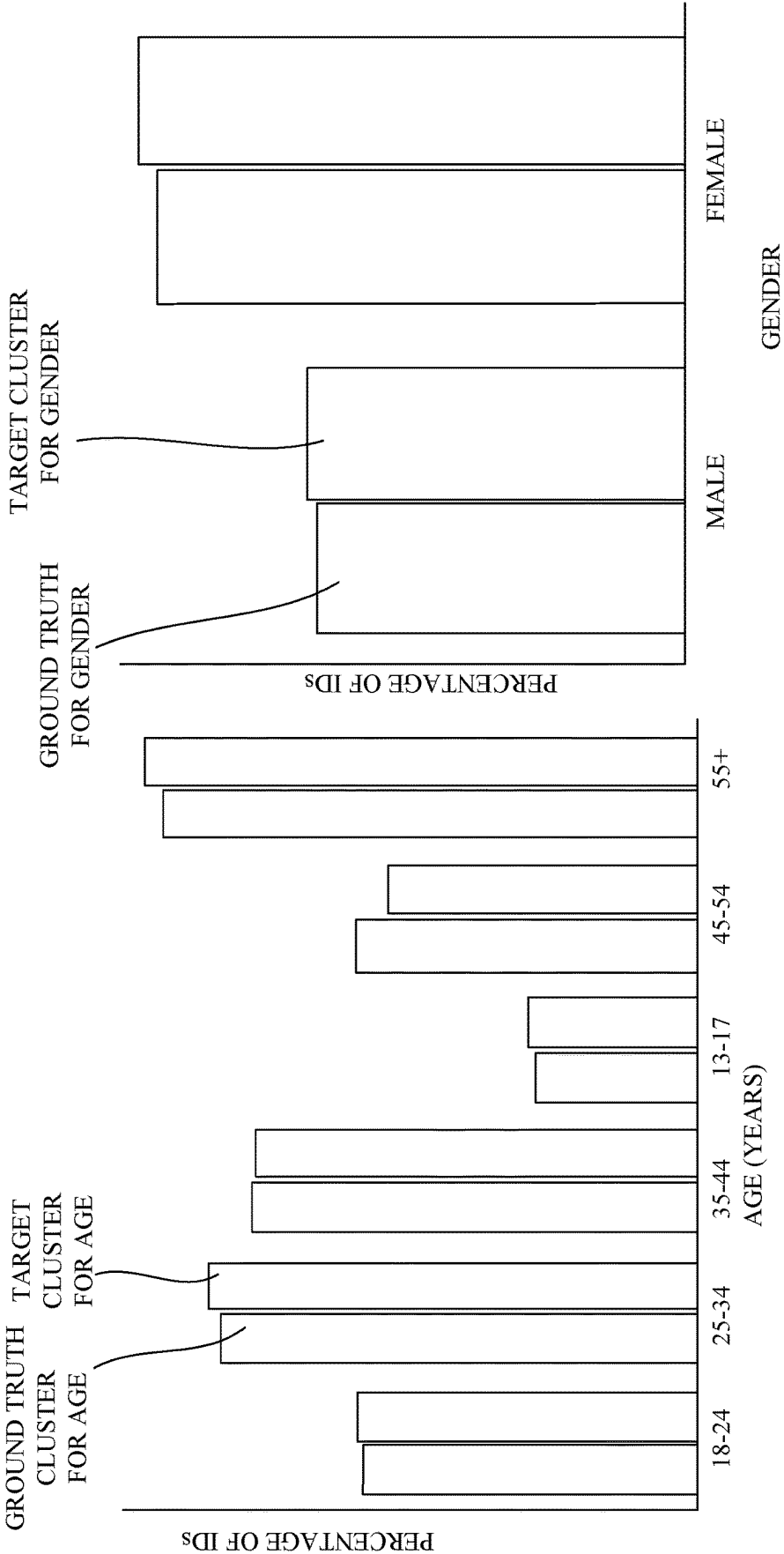


FIG. 4B

FIG. 4A

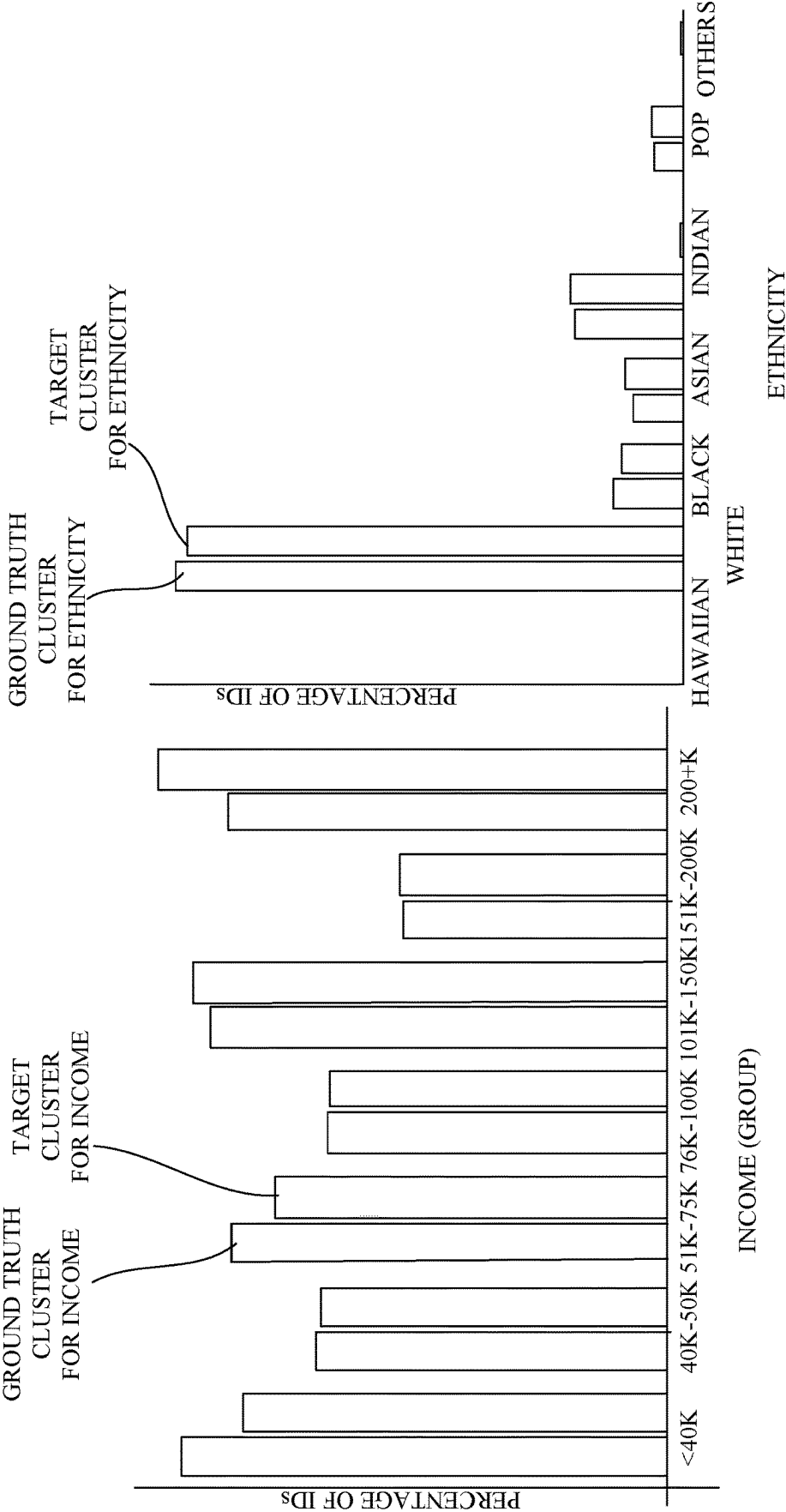
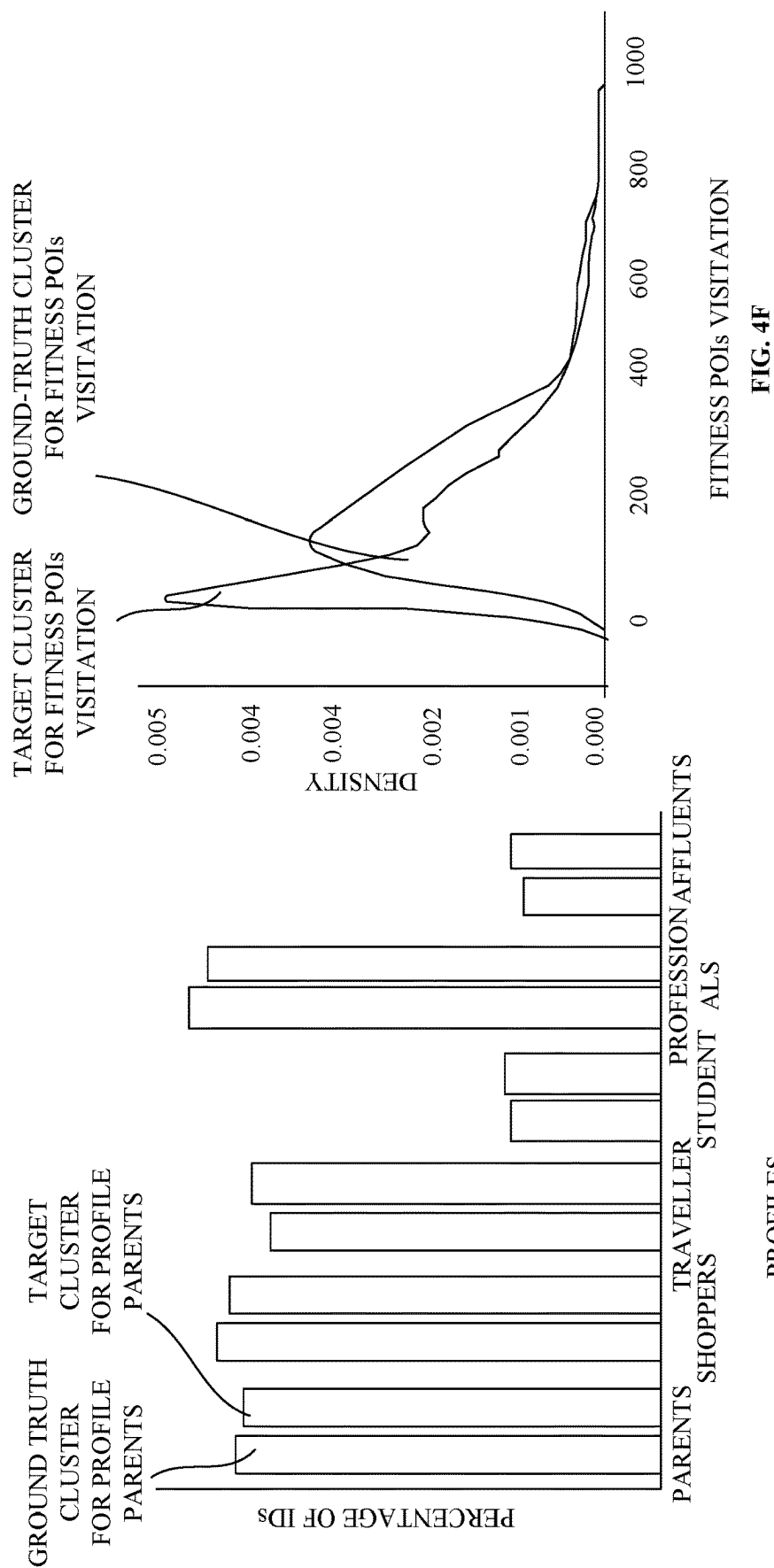


FIG. 4C

FIG. 4D



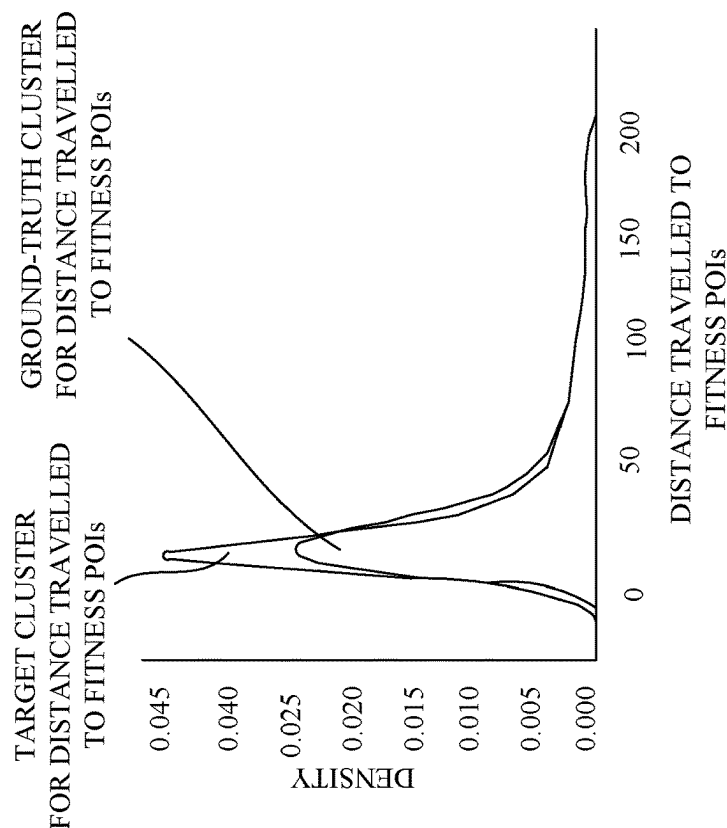


FIG. 4H

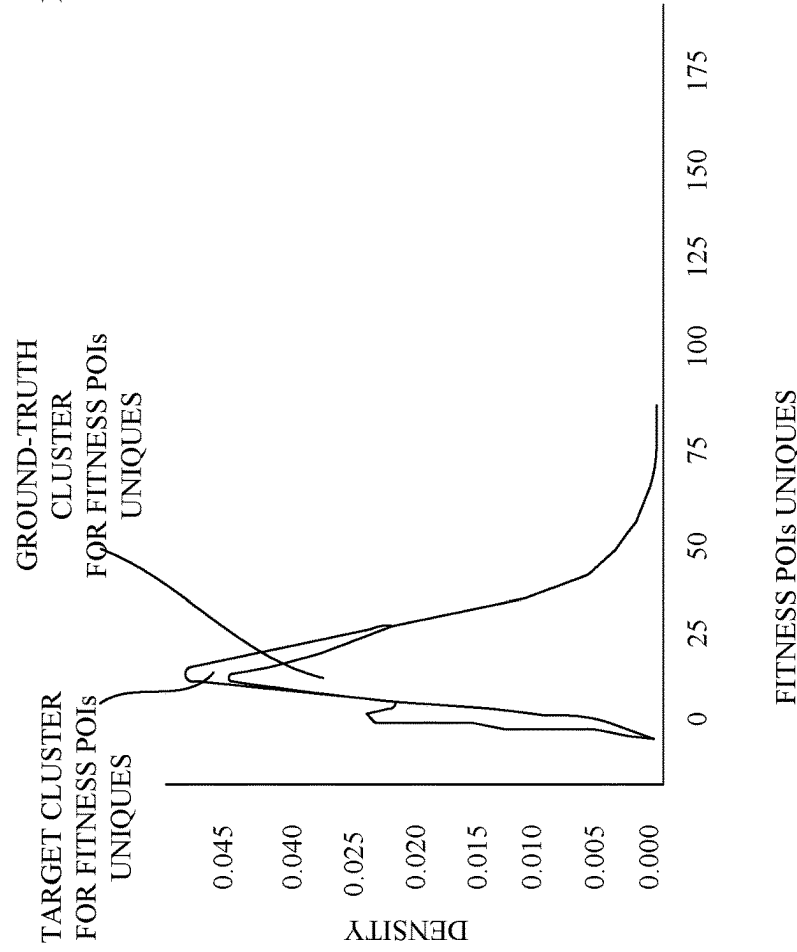


FIG. 4G

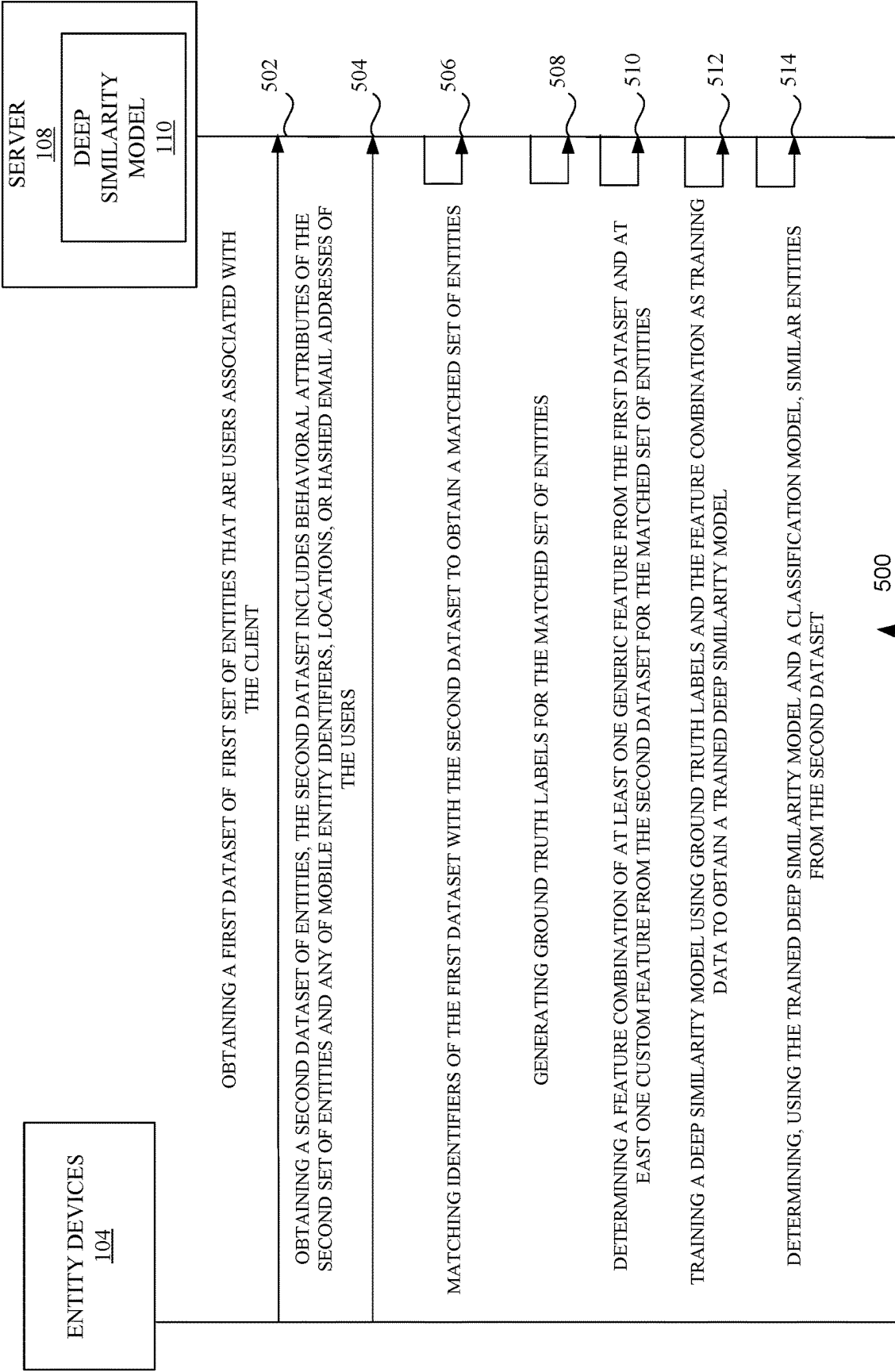


FIG. 5

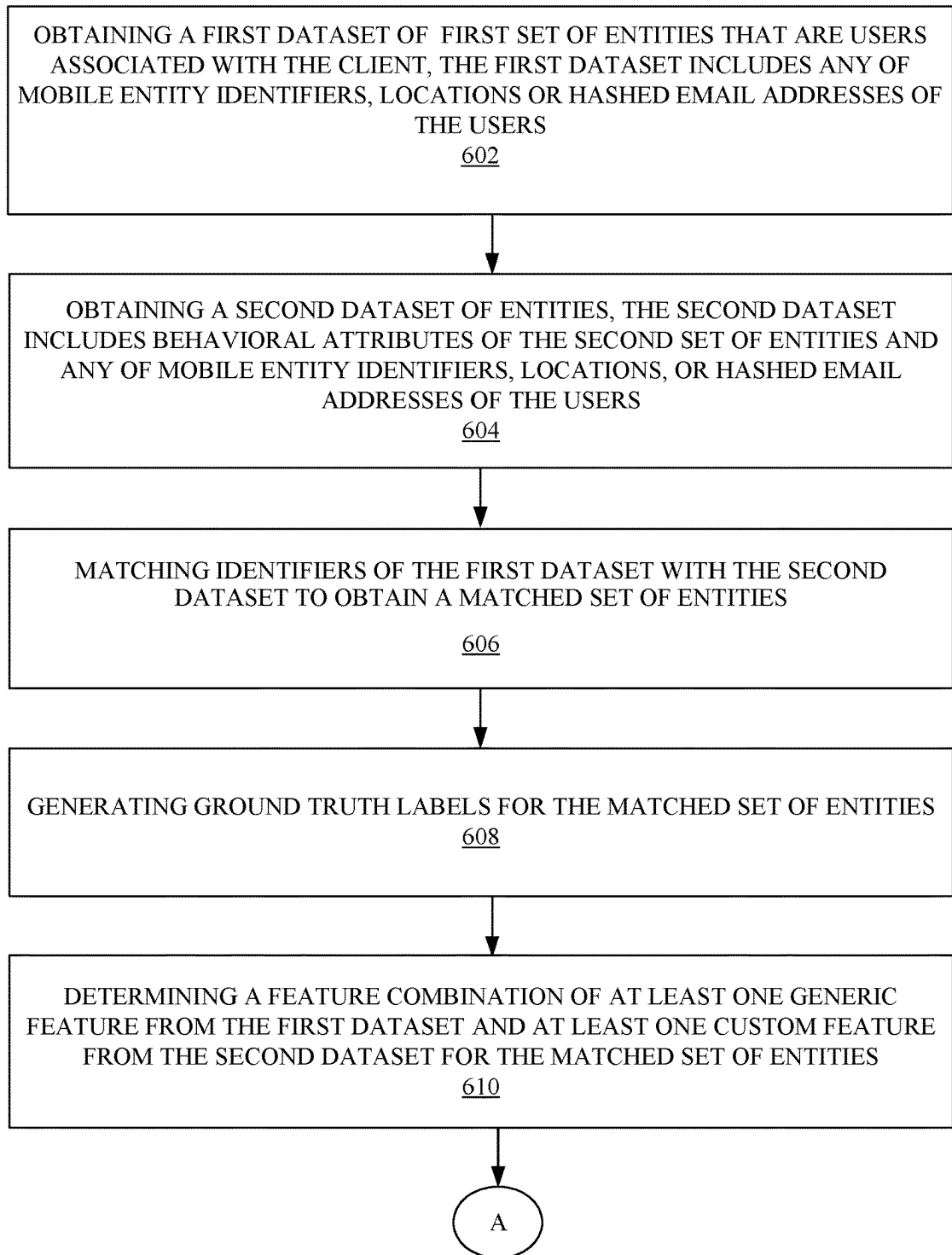


FIG. 6A

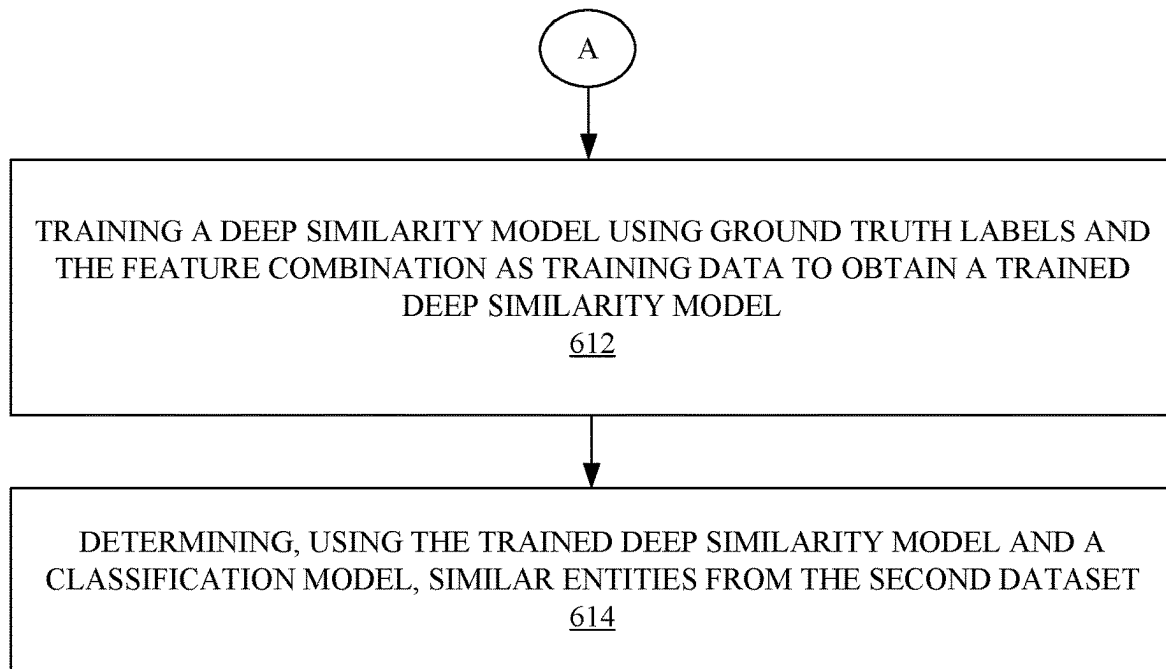


FIG. 6B

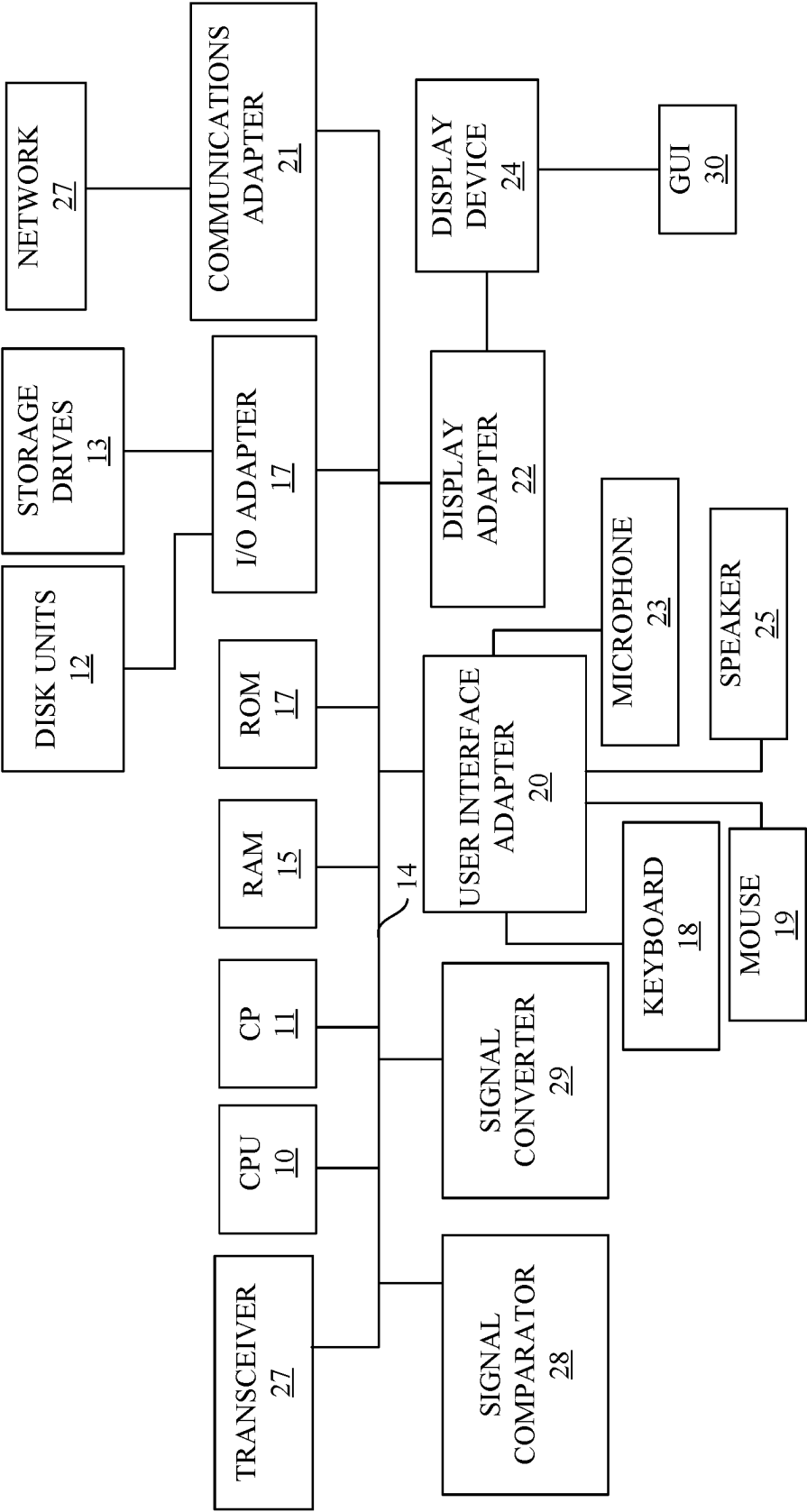


FIG. 7

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SERVER-BASED METHOD OF USING A TRAINED DEEP LEARNING MODEL AND GROUND TRUTH LABELS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/094,375, filed Jan. 8, 2023 titled "METHOD FOR PERFORMING DEEP SIMILARITY MODELLING ON CLIENT DATA TO DERIVE BEHAVIORAL ATTRIBUTES AT AN ENTITY LEVEL," the entirety of which is incorporated by reference herein.

TECHNICAL FIELD

The embodiments herein relate to deep similarity modeling, and more specifically a method for performing deep similarity modelling on client data to derive behavioral attributes at an entity level.

DESCRIPTION OF THE RELATED ART

The COVID pandemic has significantly changed behavior of the consumers to a new normal and organizations have witnessed a major upheaval in determining the behavior patterns and journey of the users of the stores. In this regard, users' vicinity shopping and dwell time in engaging with the brand has drastically changed. Henceforth, predicting a retail potential for a given store is required to get a complete picture of both people and places in a given geography.

With ever increasing digitization and usage of smart mobile applications, users are generating a large amount of internet traffic data. The internet traffic data may be an indicator of location of the users at a given time frame. A variety of different events associated with the users are encoded in a number of data formats, recorded, and transmitted in a variety of data streams depending on the nature of the device. The smart mobile applications, when engaged with a user, generate an event that produces data streams with device identifiers that are an integral part of smartphone ecosystem and smart mobile applications economy.

Further, the data streams are from independently controlled sources. The independently controlled sources are sources of the data stream that control a variety of aspects such as the attributes which are collected, frequency and means of data being collected, format of data, format of populating the data stream and types of identifiers used.

Accordingly, there remains a need to address the aforementioned technical drawbacks in existing technologies to determine behavior of the consumers in an accurate manner.

SUMMARY

In view of the foregoing, an embodiment herein provides a method for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level. The method includes (a) obtaining a first dataset of a first set of entities that are users associated with the client, the first dataset includes any of mobile entity identifiers, locations, or hashed email addresses of the users, (b) obtaining a second dataset of a second set of entities, the second dataset includes behavioral attributes of the second set of entities and any of mobile entity identifiers, locations, or hashed email addresses of the entities, (c) matching identifiers of the first dataset with the second dataset to obtain a matched set of entities, (d) generating ground truth labels for the matched

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set of entities, (e) determining a feature combination of at least one generic feature from the first dataset and at least one custom feature (specific to client) from the second dataset for the matched set of entities, (f) training a deep similarity model using ground truth labels and the feature combination as training data to obtain a trained deep similarity model, and (g) determining, using the trained deep similarity model and a classification method, similar entities from the second dataset.

In some embodiments, the method further includes (a) matching identifiers of the first dataset with the second dataset to obtain a matched set of entities, (b) generating ground truth labels for the matched set of entities, (c) determining the feature combination of the at least one generic feature from the first dataset and at least one custom feature from the second dataset for the matched set of entities, and (d) determining, using one-class classification method, similar entities from the second dataset, the similar entities are obtained when a plurality of behavioral attributes of the matched set of entities are similar to a plurality of behavioral attributes of the second set of entities while comparing each other.

In some embodiments, the method further includes (a) matching identifiers of the first dataset with the second dataset to obtain the matched set of entities, (b) determining the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities, (c) merging the feature combination with the generated ground truth labels for the matched set of entities, and (d) determining, using a binary-class classification method, contrary entities from the second dataset, the contrary entities comprise a first entity from the matched set of entities and a second entity from the second set of entities. The at least one behavioral attribute of the first entity is mutually exclusive from at least one behavioral attribute of the second entity.

In some embodiments, the method further includes (a) matching identifiers of the first dataset with the second dataset to obtain the matched set of entities, (b) generating, using classification method, ground truth labels for the matched set of entities (c) determining the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities, and (d) determining, using a multi-class classification method, entities with overlapping attributes of behavior from the second dataset, the entities with overlapping attributes of behavior are obtained when one or more behavioral attributes of the matched set of entities overlap in comparison with the plurality of behavioral attributes of the second set of entities.

In some embodiments, the method further includes merging a first behavioral attribute and a second behavioral attribute of the matched set of entities using the ground truth labels, the first behavioral attribute and the second behavioral attribute are associated with two mutually exclusive classes of behavior.

In some embodiments, the method further includes (a) obtaining weights of a plurality of behavioral attributes from the client, (b) configuring the trained deep similarity model based on the weights to obtain a re-configured model, and (c) generating a cluster for the matched set of entities using the re-configured model.

In some embodiments, the classification method depends on a level of similarity between behavioral attributes of the matched set of entities and behavioral attributes of the second set of entities.

In another aspect, there is provided a system for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level. The system includes a processor and a memory that stores a set of instructions, which when executed by the processor, causes to perform: (a) obtaining a first dataset of a first set of entities that are users associated with the client, the first dataset includes any of mobile entity identifiers, locations, or hashed email addresses of the users, (b) obtaining a second dataset of a second set of entities, the second dataset includes behavioral attributes of the second set of entities and any of mobile entity identifiers, locations, or hashed email addresses of the entities, (c) matching identifiers of the first dataset with the second dataset to obtain a matched set of entities, (d) generating, using at least one classification method, ground truth labels for the matched set of entities, (e) determining a feature combination of at least one generic feature from the first dataset and at least one custom feature (specific to client) from the second dataset for the matched set of entities, (f) training a deep similarity model using ground truth labels and the feature combination as training data to obtain a trained deep similarity model, and (g) determining, using the trained deep similarity model and a classification method, similar entities from the second dataset.

In some embodiments, the processor is configured to further include (a) matching identifiers of the first dataset with the second dataset to obtain a matched set of entities, (b) generating, using classification method, ground truth labels for the matched set of entities, (c) determining the feature combination of the at least one generic feature from the first dataset and at least one custom feature from the second dataset for the matched set of entities, and (d) determining, using one-class classification method, similar entities from the second dataset, the similar entities are obtained when a plurality of behavioral attributes of the matched set of entities are similar to a plurality of behavioral attributes of the second set of entities while comparing each other.

In some embodiments, the processor is configured to further include (a) matching identifiers of the first dataset with the second dataset to obtain the matched set of entities, (b) determining the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities, (c) merging the feature combination with the generated ground truth labels for the matched set of entities, and (d) determining, using a binary-class classification method, a combination of the similar entities and contrary entities from the second dataset, the contrary entities comprise a first entity from the matched set of entities and a second entity from the second set of entities. The at least one behavioral attribute of the first entity is mutually exclusive from at least one behavioral attribute of the second entity.

In some embodiments, the processor is configured to further include (a) matching identifiers of the first dataset with the second dataset to obtain the matched set of entities, (b) generating ground truth labels for the matched set of entities (c) determining the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities, and (d) determining, using a multi-class classification method, the similar entities of multiple overlapping attributes of behavior from the second dataset, the similar entities of multiple overlapping attributes of behavior are obtained when one or more behavioral attributes of

the matched set of entities overlap in comparison with the one or more behavioral attributes of the second set of entities.

In some embodiments, the processor is configured to further include merging a first behavioral attribute and a second behavioral attribute of the matched set of entities using the ground truth labels, the first behavioral attribute, and the second behavioral attribute are associated with two mutually exclusive classes of behavior.

In some embodiments, the processor is configured to further include (a) obtaining weights of one or more behavioral attributes from the client, (b) configuring the trained deep similarity model based on the weights to obtain a re-configured model, and (c) generating a cluster for the matched set of entities using the re-configured model.

In some embodiments, the classification method depends on a level of similarity between behavioral attributes of the matched set of entities and behavioral attributes of the second set of entities.

In another aspect, there is provided one or more non-transitory computer-readable storage mediums storing the one or more sequences of instructions, which when executed by the one or more processors, causes performing a deep similarity modeling on client data to derive behavioral attributes at an entity level by (a) obtaining a first dataset of a first set of entities that are users associated with the client, the first dataset includes any of mobile entity identifiers, locations, or hashed email addresses of the users, (b) obtaining a second dataset of a second set of entities, the second dataset includes behavioral attributes of the second set of entities and any of mobile entity identifiers, locations, or hashed email addresses of the entities, (c) matching identifiers of the first dataset with the second dataset to obtain a matched set of entities, (d) generating ground truth labels for the matched set of entities, (e) determining a feature combination of at least one generic feature from the first dataset and at least one custom feature (specific to client) from the second dataset for the matched set of entities, (f) training a deep similarity model using ground truth labels and the feature combination as training data to obtain a trained deep similarity model, and (g) determining, using the trained deep similarity model and a classification method, similar entities from the second dataset.

In some embodiments, the sequence of instructions further includes (a) matching identifiers of the first dataset with the second dataset to obtain a matched set of entities, (b) generating ground truth labels for the matched set of entities, (c) determining the feature combination of the at least one generic feature from the first dataset and at least one custom feature from the second dataset for the matched set of entities, and (d) determining, using one-class classification method, similar entities from the second dataset, the similar entities are obtained when a plurality of behavioral attributes of the matched set of entities are similar to one or more behavioral attributes of the second set of entities while comparing each other.

In some embodiments, the sequence of instructions further includes (a) matching identifiers of the first dataset with the second dataset to obtain the matched set of entities, (b) determining the feature combination of the at least one generic feature from the first dataset, and the at least one custom feature from the second dataset for the matched set of entities, (c) merging the feature combination with the generated ground truth labels for the matched set of entities, and (d) determining, using a binary-class classification method, a combination of the similar entities and contrary entities from the second dataset, the contrary entities com-

prise a first entity from the matched set of entities and a second entity from the second set of entities. The at least one behavioral attribute of the first entity is mutually exclusive from at least one behavioral attribute of the second entity.

In some embodiments, the sequence of instructions further includes merging a first behavioral attribute and a second behavioral attribute of the matched set of entities using the ground truth labels, the first behavioral attribute, and the second behavioral attribute are associated with two mutually exclusive classes of behavior.

In some embodiments, the sequence of instructions further includes (a) obtaining weights of a plurality of behavioral attributes from the client, (b) configuring the trained deep similarity model based on the weights to obtain a re-configured model, and (c) generating a cluster for the matched set of entities using the re-configured model.

In some embodiments, the classification method depends on a level of similarity between behavioral attributes of the matched set of entities and behavioral attributes of the second set of entities.

A system and method for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level are provided. The system provides a scalable model at user ID level scoring. Thereby, behavioral attributes of entities are achieved. Hence, user clusters with a high confidence level are achieved with sample ingestion. The system enables visibility of any product's brand.

These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following descriptions, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments herein will be better understood from the following detailed description with reference to the drawings, in which:

FIG. 1 is a schematic illustration of a system for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level according to some embodiments herein;

FIG. 2 is a block diagram of a server of FIG. 1 according to some embodiments herein;

FIG. 3A is an exemplary flow diagram of performing a deep similarity modeling on client data to derive behavioral attributes using a one-class classification method, according to some embodiments herein;

FIG. 3B is an exemplary flow diagram of performing a deep similarity modeling on client data to derive behavioral attributes using a binary classification method, according to some embodiments herein;

FIG. 3C is an exemplary flow diagram of performing a deep similarity modeling on client data to derive behavioral attributes using a multi-class classification method, according to some embodiments herein;

FIG. 4A is a graphical representation of user clusters based on an age group that illustrates ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 4B is a graphical representation of user clusters based on gender that illustrates ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 4C is a graphical representation of user clusters based on income that illustrates ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 4D is a graphical representation of user clusters based on ethnicity that illustrates ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 4E is a graphical representation of user clusters based on profiles that illustrate ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 4F is a graphical representation of user clusters based on fitness visitations that illustrate ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 4G is a graphical representation of user clusters based on fitness uniques that illustrate ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 4H is a graphical representation of user clusters based on distance travelled to fitness centers that illustrates ground-truth clusters vs target clusters of one or more entities, according to some embodiments herein;

FIG. 5 illustrates an interaction diagram of a method for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level according to some embodiments herein;

FIGS. 6A and 6B are flow diagrams of a method for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level according to some embodiments herein; and

FIG. 7 is a schematic diagram of a computer architecture of the unique generated identifier server or one or more devices in accordance with embodiments herein.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.

There remains a need for a system and method for performing a deep similarity modeling, and more specifically, for an automatic system and method for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level. Referring now to the drawings, and more particularly to FIGS. 1 to 7, where similar reference characters denote corresponding features consistently throughout the figures, there are shown preferred embodiments.

The term "independently controlled data sources" refers to any source that may control or standardize different

aspects of data streams. The different aspects include, but are not limited to, (1) a type of data that needs to be collected, (2) a time and location of the data that needs to be collected, (3) a data collection method, (4) modification of collected data, (5) a portion of data to be revealed to the public, (6) a portion of the data to be protected, (7) a portion of data that can be permitted by a consumer or a user of an application or the device, and (8) a portion of data to be completely private. The terms “consumer” and “user” may be used interchangeably and refer to an entity associated with a network device or an entity device.

A single real-world event may be tracked by different independently controlled data sources. Alternatively, data from the different independently controlled data sources may be interleaved to understand an event or a sequence of events. For example, consider the consumer using multiple applications on his smartphone, as he or she interacts with each application, multiple independent data streams of the sequence of events may be produced. Each application may become an independent data source. Events and users may have different identifiers across different applications depending on how the application is implemented. Additionally, if one were to monitor a network, each application-level event may generate additional lower-level network events.

In an exemplary embodiment, various modules described herein and illustrated in the figures are embodied as hardware-enabled modules and may be configured as a plurality of overlapping or independent electronic circuits, devices, and discrete elements packaged onto a circuit board to provide data and signal processing functionality within a computer. An example might be a comparator, inverter, or flip-flop, which could include a plurality of transistors and other supporting devices and circuit elements. The modules that are configured with electronic circuits process computer logic instructions capable of providing at least one digital signal or analog signal for performing various functions as described herein.

FIG. 1 is a schematic illustration of a system **100** for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level according to some embodiments herein. The system **100** includes one or more entity devices **104A-N** associated with one or more entities **102A-N**, and a server **108**. The one or more entity devices **104A-N** include one or more smart mobile applications. The one or more entity devices **104A-N** are communicatively connected to the server **108** through a network **106**. In some embodiments, the network **106** is at least one of a wired network, a wireless network, a combination of the wired network and the wireless network or the Internet.

In some embodiments, the one or more entity devices **104A-N** include, but are not limited to, a mobile device, a smartphone, a smartwatch, a notebook, a Global Positioning System (GPS) device, a tablet, a desktop computer, a laptop or any network-enabled device that generates the location data streams.

The server **108** obtains the first dataset of the first set of entities. The first set of entities are entities that are associated with the client. The first dataset includes any mobile entity identifiers, locations, cookies, or hashed email addresses of the users. The server **108** obtains the second dataset of the second set of entities. The second dataset includes behavioral attributes of the second set of entities and any mobile entity identifiers, locations, or hashed email addresses of the entities.

The second set of entities may be user attributes, financial data, offline behavior, online behavior, social media, etc. The

user attributes may include but are not limited to, demographics like gender, age group, income, ethnicity, profiles like parents, professionals, shoppers, travelers, affluents, health conscious, foodies, home location or proximity from home to store, dwell time at a store, brand affinity. The financial data may include, but is not limited to, point of sale like transaction date, long visits to a POI online/offline, size of a wallet, and share of wallet. The offline behavior may include, but is not limited to, location using probabilistic ping to POI assignment algorithm. The online behavior may include, but is not limited to, browsing habits like websites, articles, and products. Social media may include, but is not limited to, likability/dislike for some products, and purchase intent.

The server **108** may be configured to obtain the first dataset and the second dataset by location mapping of the one or more entities **102A-N**. The server **108** may be configured to generate, using one or more location data streams that are associated with the one or more entities **102A-N**, a location mapping of the one or more entities **102A-N** with a geographical area. The location mapping may provide an ambient population of the geographical area of the one or more entities **102A-N**. The one or more location data streams may be obtained from independently controlled data sources. The location data streams may include a real-time event with additional information including device attributes, connection attributes, and user agent strings. A connection attribute is a connection-indicative signal that may be generated at the one or more entity devices **104A-N**. The connection attribute may be indicative of a presence or a characteristic of a connection between the one or more entity devices **104A-N** and at least one other entity device of the one or more entity devices **104A-N** or a server. The one or more connection attributes may include, but not be limited to, a connection type, an internet protocol address, and a carrier. For example, the one or more connection attributes may be “Cell4g,203.218.177.24,454-00”. The user agent strings contain a number of tokens that refer to various aspects of a request from the one or more entity devices **104A-N** to the server **108**, including a browser name and a browser version, a rendering engine, the model number attribute of the one or more entity devices **104A-N**, the operating system. For example, the user agent strings may be (a) “Mozilla/5.0 (Linux; Android 6.0; S9_N Build/MRA58K; wv)”, (b) “AppleWebKit/537.36 (KHTML, like Gecko) Version/4.0”, (c) “Chrome/84.0.4147.125” and (d) “Mobile Safari/537.36”. Engagement of the one or more entity devices **104A-N** with wi-fi hotspots may be tracked using location data streams that may be obtained from the different independently controlled data sources which may include telecom operators or smart mobile application data aggregators. The location data stream is the event or the sequence of events associated with time and location (longitude and latitude) and may also include additional payload information. The event or the sequence of events may be tracked by the different independently controlled data sources. For example, consider an entity **102A** or a user using one or more smart mobile applications on an android phone associated with the entity **102A**. As he or she interacts with each application, multiple independent streams of events may be produced and each application becomes an independent data source. Events and the one or more entity devices **104A-N** may have different identifiers across different applications depending on how the smart mobile application is implemented. Additionally, if the network **106** were to be monitored, each smart mobile application-level event may generate additional lower-level network events.

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The term “location” refers to a geographic location that includes a latitude-longitude pair and/or an altitude. The location may include a locality, a sub locality, an establishment, a geocode or an address. The location may be any geographic location on land or sea.

In some embodiments, the one or more entity devices **104A-N** may run the one or more smart mobile applications that are responsible to generate location data streams.

In some embodiments, the independently controlled data sources may include (a) real-time bidding data that is an incoming data source that may be used for targeting an entity, (b) software development kit data that provides increased control, accuracy, and trust in the location data streams, and (c) third-party data sources that include app graph and professional data that may be used to enrich and build device signatures, or a list of normalized device models.

The server **108** may be configured to match identifiers of the first dataset with the second dataset to obtain a matched set of entities. The server **108** may be configured to generate ground truth labels for the matched set of entities using high confident entities. The ground truth labels for the matched set of entities may be also known as profiles. For example, the following tables (table 1, table 2, table 3) provide different profiles of entities.

TABLE 1

Name	Online client	Offline client	Digitally active client	Home location closer to store	Experimental mindset	Intent towards competitor	Student	Age	Score
John	Yes	No	Yes	Yes	High	Low	Yes	18-24	High Value

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TABLE 2

Name	Online client	Offline client	Digitally active client	Home location closer to store	Experimental mindset	Intent towards competitor	Student	Age	Score
Mary	No	Yes	No	No (Far off)	High	Low	Yes	45+	Low Value

TABLE 3

Name	Online client	Offline client	Digitally active client	Home location closer to store	Experimental mindset	Intent towards competitor	Student	Age	Score
Kate	Yes	Yes	Yes	Yes	High	High	Yes	18-24	High Value

The server **108** may be configured to determine a feature combination of at least one generic feature from the first dataset and at least one custom feature (specific to client) from the second dataset for the matched set of entities. The server **108** may be configured to train a deep similarity model **110** using ground truth labels and the feature combination as training data to obtain a trained deep similarity model.

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The server **108** may be configured to determine similar entities from the second dataset using the trained deep similarity model and a classification method.

In some embodiments, the method further includes merging a first behavioral attribute and a second behavioral attribute of the matched set of entities using the ground truth labels, the first behavioral attribute and the second behavioral attribute are associated with two mutually exclusive classes of behavior.

In some embodiments, the method further includes (a) obtaining weights of one or more behavioral attributes from the client, (b) configuring the trained deep similarity model based on the weights to obtain a re-configured model, and (c) generating a cluster for the matched set of entities using the re-configured model. The following table 4 depicts an exemplary generation of a cluster for the matched set of entities, for example, fitness enthusiasts, based on the weights of one or more behavioral attributes against the entities, for example, low or medium.

TABLE 4

Encrypted Device ID	fitness enthusiast
7afc56283a18723a6ab43aa540267c31	low
188728d90e709816663d60db1bae62b9	medium

TABLE 4-continued

Encrypted Device ID	fitness enthusiast
3022c2dfbdb856f6f01f7af107070396	medium
620f5317d7f7469f13d0e50976b3efc4	medium
3f0243688c0cea34df60a791369586a7	medium
44a385723b0e0fd2d2579e5c39c0c540	medium

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In some embodiments, the classification method depends on a level of similarity between behavioral attributes of the matched set of entities and behavioral attributes of the second set of entities.

FIG. 2 is a block diagram of the server 108 of FIG. 1 according to some embodiments herein. The server 108 includes a database 202, a first dataset obtaining module 204, a second dataset obtaining module 206, identifiers matching module 208, a ground truth labels generating module 210, a feature combination determining module 212, the deep similarity model 110 and similar entities determining module 214. The database 202 stores the first dataset, and the second dataset. The first dataset and the second dataset include the one or more location data streams that are obtained from independently controlled data sources where the location data streams include a real-time event with additional information including device attributes, connection attributes, user agent strings, behavioral attributes, mobile entity identifiers, locations, or hashed email addresses of the entities.

The first dataset obtaining module 204 is configured to obtain the first dataset of the first set of entities that are users associated with the client. The first dataset includes any of the mobile entity identifiers, locations, or hashed email addresses of the users.

The second dataset obtaining module 206 is configured to obtain the second dataset of the second set of entities. The second dataset includes behavioral attributes of the second set of entities and any mobile entity identifiers, locations, or hashed email addresses of the entities.

The identifiers matching module 208 is configured to match identifiers of the first dataset with the second dataset to obtain a matched set of entities. The ground truth labels generating module 210 is configured to generate ground truth labels for the matched set of entities using high confident entities.

The feature combination determining module 212 is configured to determine a feature combination of at least one generic feature from the first dataset and at least one custom feature (specific to the client) from the second dataset for the matched set of entities. The deep similarity model 110 is trained using ground truth labels and the feature combination as training data to obtain a trained deep similarity model.

The similar entities determining module 214 is configured to determine similar entities from the second dataset using the trained deep similarity model and a classification method.

FIG. 3A is an exemplary flow diagram of performing a deep similarity modeling on client data to derive behavioral attributes using a one-class classification method, according to some embodiments herein. The exemplary flow diagram includes matching, using the identifiers matching module 208, identifiers of the first dataset with the second dataset to obtain the matched set of entities. The exemplary flow diagram includes determining, using the feature combination determining module 212, the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities. The at least one generic feature from the first dataset is determined by generic features determining module 302. The at least one custom feature from the second dataset is determined by custom features determining module 304. The exemplary flow diagram includes merging the feature combination with the generated ground truth labels for the matched set of entities. The exemplary flow diagram includes determining, using the one-class classification

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module 306, similar entities from the second dataset, the similar entities are obtained when one or more behavioral attributes of the matched set of entities are similar to one or more behavioral attributes of the second set of entities while comparing each other.

FIG. 3B is an exemplary flow diagram of performing a deep similarity modeling on client data to derive behavioral attributes using a binary classification method, according to some embodiments herein. The exemplary flow diagram includes matching, using the identifiers matching module 208, identifiers of the first dataset with the second dataset to obtain the matched set of entities. The exemplary flow diagram includes determining, using the feature combination determining module 212, the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities. The at least one generic feature from the first dataset is determined by generic features determining module 302. The at least one custom feature from the second dataset is determined by custom features determining module 304. The exemplary flow diagram includes merging the feature combination with the generated ground truth labels for the matched set of entities. The exemplary flow diagram includes determining, using the binary class classification module 308, a combination of the similar entities and contrary entities from the second dataset, the contrary entities comprise the first entity from the matched set of entities and the second entity from the second set of entities. The at least one behavioral attribute of the first entity is mutually exclusive from at least one behavioral attribute of the second entity.

FIG. 3C is an exemplary flow diagram of performing a deep similarity modeling on client data to derive behavioral attributes using a multi-class classification method, according to some embodiments herein. The exemplary flow diagram includes matching, using the identifiers matching module 208, identifiers of the first dataset with the second dataset to obtain the matched set of entities. The exemplary flow diagram includes determining, using the feature combination determining module 212, the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities. The at least one generic feature from the first dataset is determined by generic features determining module 302. The at least one custom feature from the second dataset is determined by custom features determining module 304. The exemplary flow diagram includes determining, using the multi-class classification module 310, the similar entities of multiple overlapping attributes of behavior from the second dataset, the similar entities of multiple overlapping attributes of behavior are obtained when one or more behavioral attributes of the matched set of entities overlap in comparison with the plurality of behavioral attributes of the second set of entities.

FIG. 4A is a graphical representation of user clusters based on an age group that illustrates ground-truth clusters vs target clusters of one or more entities 102A-N, according to some embodiments herein. The graphical representation depicts the percentage of user IDs on the Y axis and the age group in years on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of the one or more entities 102A-N based on age groups.

FIG. 4B is a graphical representation of user clusters based on gender that illustrates ground-truth clusters vs target clusters of one or more entities 102A-N, according to some embodiments herein. The graphical representation depicts the percentage of user IDs on the Y axis and gender

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on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of one or more entities **102A-N** based on gender.

FIG. **4C** is a graphical representation of user clusters based on income that illustrates ground-truth clusters vs target clusters of one or more entities **102A-N**, according to some embodiments herein. The graphical representation depicts the percentage of user IDs on the Y axis and the income group on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of one or more entities **102A-N** based on income group.

FIG. **4D** is a graphical representation of user clusters based on ethnicity that illustrates ground-truth clusters vs target clusters of one or more entities **102A-N**, according to some embodiments herein. The graphical representation depicts the percentage of user IDs on the Y axis and ethnicity on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of one or more entities **102A-N** based on ethnicity.

FIG. **4E** is a graphical representation of user clusters based on profiles that illustrate ground-truth clusters vs target clusters of one or more entities **102A-N**, according to some embodiments herein. The graphical representation depicts the percentage of user IDs on the Y axis and profiles on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of one or more entities **102A-N** based on profiles.

FIG. **4F** is a graphical representation of user clusters based on fitness visitations that illustrate ground-truth clusters vs target clusters of one or more entities **102A-N**, according to some embodiments herein. The graphical representation depicts the percentage of density on the Y axis and fitness visitations on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of one or more entities **102A-N** based on fitness visitations.

FIG. **4G** is a graphical representation of user clusters based on fitness uniques that illustrate ground-truth clusters vs target clusters of one or more entities **102A-N**, according to some embodiments herein. The graphical representation depicts the percentage of density on the Y axis and fitness uniques on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of one or more entities **102A-N** based on fitness uniques.

FIG. **4H** is a graphical representation of user clusters based on distance travelled to fitness centers that illustrates ground-truth clusters vs target clusters of one or more entities **102A-N**, according to some embodiments herein. The graphical representation depicts the percentage of density on the Y axis and the distance travelled to fitness centers on the X axis. The graphical representation depicts ground-truth clusters vs target clusters of one or more entities **102A-N** based on the distance travelled to fitness centers.

FIG. **5** illustrates an interaction diagram **500** of a method for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level according to some embodiments herein. At step **502**, a first dataset of a first set of entities that are users associated with the client are obtained. At step **504**, a second dataset of a second set of entities are obtained. At step **506**, identifiers of the first dataset are matched with the second dataset to obtain a matched set of entities. At step **508**, ground truth labels for the matched set of entities are generated. The matched set of entities are generated using high confident entities. At step **510**, a feature combination of at least one generic feature from the first dataset and at least one custom feature (specific to client) from the second dataset for the matched set of entities are determined. At step **512**, a deep similarity model

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is trained using ground truth labels and the feature combination as training data to obtain a trained deep similarity model. At step **514**, similar entities from the second dataset are determined using the trained deep similarity model and a classification method.

FIGS. **6A** and **6B** are flow diagrams of a method for performing a deep similarity modeling on client data to derive behavioral attributes at an entity level according to some embodiments herein. At step **602**, the method includes obtaining a first dataset of a first set of entities that are users associated with the client. The first dataset includes any of mobile entity identifiers, locations, or hashed email addresses of the users. At step **604**, the method includes obtaining a second dataset of a second set of entities. The second dataset includes behavioral attributes of the second set of entities and any of mobile entity identifiers, locations, or hashed email addresses of the entities. At step **606**, the method includes matching identifiers of the first dataset with the second dataset to obtain a matched set of entities. At step **608**, the method includes generating ground truth labels for the matched set of entities. The matched set of entities are generated using high confident entities. At step **610**, the method includes determining a feature combination of at least one generic feature from the first dataset and at least one custom feature (specific to client) from the second dataset for the matched set of entities. At step **612**, the method includes training a deep similarity model using ground truth labels and the feature combination as training data to obtain a trained deep similarity model. At step **614**, the method includes determining, using the trained deep similarity model and a classification method, similar entities from the second dataset.

In some embodiments, the processor is configured to further include (a) matching identifiers of the first dataset with the second dataset to obtain a matched set of entities, (b) generating ground truth labels for the matched set of entities, (c) determining the feature combination of the at least one generic feature from the first dataset and at least one custom feature from the second dataset for the matched set of entities, and (d) determining, using one-class classification method, similar entities from the second dataset, the similar entities are obtained when a plurality of behavioral attributes of the matched set of entities are similar to a plurality of behavioral attributes of the second set of entities while comparing each other.

In some embodiments, the processor is configured to further include (a) matching identifiers of the first dataset with the second dataset to obtain the matched set of entities, (b) determining the feature combination of the at least one generic feature from the first dataset and the at least one custom feature from the second dataset for the matched set of entities, (c) merging the feature combination with the generated ground truth labels for the matched set of entities, and (d) determining, using a binary-class classification method, a combination of the similar entities and contrary entities from the second dataset, the contrary entities comprise a first entity from the matched set of entities and a second entity from the second set of entities. The at least one behavioral attribute of the first entity is mutually exclusive from at least one behavioral attribute of the second entity.

In some embodiments, the processor is configured to further include merging a first behavioral attribute and a second behavioral attribute of the matched set of entities using the ground truth labels, the first behavioral attribute, and the second behavioral attribute are associated with two mutually exclusive classes of behavior.

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In some embodiments, the processor is configured to further include (a) obtaining weights of one or more behavioral attributes from the client, (b) configuring the trained deep similarity model based on the weights to obtain a re-configured model, and (c) generating a cluster for the matched set of entities using the re-configured model.

In some embodiments, the classification method depends on a level of similarity between behavioral attributes of the matched set of entities and behavioral attributes of the second set of entities.

A representative hardware environment for practicing the embodiments herein is depicted in FIG. 7, with reference to FIGS. 1 through 6A and 6B. This schematic drawing illustrates a hardware configuration of a server 108 or a computer system or a computing device in accordance with the embodiments herein. The system includes at least one processing device CPU 10 that may be interconnected via system bus 14 to various devices such as a random-access memory (RAM) 12, read-only memory (ROM) 16, and an input/output (I/O) adapter 18. The I/O adapter 18 can connect to peripheral devices, such as disk units 38 and program storage devices 40 that are readable by the system. The system can read the inventive instructions on the program storage devices 40 and follow these instructions to execute the methodology of the embodiments herein. The system further includes a user interface adapter 22 that connects a keyboard 28, mouse 30, speaker 32, microphone 34, and other user interface devices such as a touch screen device (not shown) to the bus 14 to gather user input. Additionally, a communication adapter 20 connects the bus 14 to a data processing network 42, and a display adapter 24 connects the bus 14 to a display device 26, which provides a graphical user interface (GUI) 36 of the output data in accordance with the embodiments herein, or which may be embodied as an output device such as a monitor, printer, or transmitter, for example.

The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the spirit and scope.

What is claimed is:

1. A server-implemented method for determining, at a server, using a trained deep learning model, a cluster of device identifiers associated with computing devices having truth labels based on locations data streams obtained from the computing devices, the method comprising:

obtaining, at the server, a first dataset of a first set of entities, each entity of the first data set being a user associated with a client computing device, wherein the first dataset comprises entity identifiers, locations, cookies, and hashed email addresses of the entities;

obtaining, at the server, a second dataset of a second set of entities, each entity associated with a client computing device in a geographical area, wherein the second dataset comprises behavioral attributes of the second set of computing devices, mobile entity identifiers,

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hashed email addresses of the entities, and the locations data streams, the locations data streams comprising characteristics, connection characteristics, and user agent strings, the user agent strings comprising a plurality of tokens comprising a request from the second set of computing devices;

generating, at the server, a plurality of match sets of entities, each match set comprising a first client computing device associated with an entity of the first dataset and a second client computing device associated with an entity of the second dataset;

determining, at the server, for each of the plurality of match sets, a ground truth label, each ground truth label determined using at least high confidence entities of the first dataset and the second dataset;

providing, at the server, to the trained deep learning model, the ground truth labels;

using the trained deep learning model to generate the cluster of device identifiers, the cluster of device identifiers generated based on a level of similarity between behavioral attributes of the match set and behavioral attributes of the second set of computing devices, each device identifier of the cluster being associated with one of the computing devices associated with the entities of the second dataset; and

obtaining, at the server, from the trained deep learning model, the cluster, the cluster comprising a plurality of device identifiers and a behavioral attribute for each device identifier.

2. The server-implemented method of claim 1, wherein the trained deep learning model is trained using the ground truth labels and at least one custom feature specific to the client.

3. The server-implemented method of claim 1, further comprising: determining, using the trained deep learning model and a one-class classification method on the second dataset, the cluster of the device identifiers associated with the computing devices having the characteristics that are similar to the match set from the first dataset, from the second dataset.

4. The server-implemented method of claim 1, wherein at least one characteristic of the first computing device is mutually exclusive from at least one characteristic of the second computing device.

5. The server-implemented method of claim 4, further comprising: merging a first behavioral attribute and a second behavioral attribute of the matched set using the ground truth labels, wherein the first behavioral attribute and the second behavioral attribute are associated with two mutually exclusive classes of behavior.

6. The server-implemented method of claim 1, further comprising: determining, using the trained deep learning model and a classification method, the cluster of the device identifiers having multiple overlapping characteristics of behavior to the confident computing devices from the first dataset, from the second dataset, wherein the device identifiers having the multiple overlapping characteristics of behavior are obtained when a plurality of first behavioral characteristics of the matched set overlap in comparison with a plurality of second behavioral characteristics of the second set.

7. The server-implemented method of claim 1, further comprising: scoring the matched set of computing devices against a behavioral attribute by: generating a user scoring model based on a function of the behavioral attributes of the matched set of computing entities; and assigning, using the

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user scoring model, a score for each of the matched set of computing entities against the behavioral attribute.

8. The server-implemented method of claim 1, further comprising: obtaining weights of a plurality of behavioral attributes from the client; configuring the trained deep learning model based on the weights to obtain a reconfigured deep learning model; and generating a cluster for a matched set of computing devices using the re-configured deep learning model.

9. The server-implemented method of claim 1, wherein the trained machine learning model employs a classification method, the classification method depending on a level of learning between behavioral attributes of the matched set and behavioral attributes of the second set.

10. A system for determining, at a server, using a trained deep learning model, a cluster of device identifiers associated with computing devices having characteristics based on locations data streams obtained from the computing devices, the system comprising:

a processor; and

a memory that stores a set of instructions, which when executed by the processor, causes it to perform:

obtaining, at the server, a first dataset of a first set of computing entities, each entity of the first data set being a user associated with a client computing device, wherein the first dataset comprises entity identifiers, locations, cookies, and hashed email addresses of the entities;

obtaining, at the server, a second dataset of a second set of entities, each entity associated with a client computing device in a geographical area, wherein the second dataset comprises behavioral attributes of the second set of computing devices, mobile entity identifiers, hashed email addresses of the entities, and the locations data streams, the location data streams comprising characteristics, connection characteristics, and user agent strings, the user agent strings comprising a plurality of tokens comprising a request from the second set of computing devices;

generating, at the server, a plurality of match sets of entities, each match set comprising a first client computing device associated with an entity of the first dataset and a second client computing device associated with an entity of the second dataset;

determining, at the server, for each of the plurality of match sets, a ground truth label, each ground truth label determined using at least high confidence entities of the first dataset and the second dataset;

providing, at the server, to the trained deep learning model, the ground truth labels; using the trained deep learning model to generate the cluster of device identifiers, the cluster of device identifiers generated based on a level of similarity between behavioral attributes of the match set and behavioral attributes of the second set of computing devices, each device identifier of the

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cluster being associated with one of the computing devices associated with the entities of the second dataset; and

obtaining, at the server, from the trained deep learning model, the cluster, the cluster comprising a plurality of device identifiers and a behavioral attribute for each device identifier.

11. The system of claim 10, wherein the processor further performs determining, using the trained deep learning model and a one-class classification method on the second dataset, the cluster of the device identifiers associated with the computing devices of the computing devices having the characteristics that are similar to the confident computing devices from the first dataset, from the second dataset.

12. The system of claim 10, wherein at least one characteristic of the first computing device is mutually exclusive from at least one characteristic of the second computing device.

13. The system of claim 12, wherein the processor further performs merging a first behavioral attribute and a second behavioral attribute of the matched set using the ground truth labels, wherein the first behavioral attribute and the second behavioral attribute are associated with two mutually exclusive classes of behavior.

14. The system of claim 10, wherein the processor further performs determining, using the trained deep learning model and a classification method, the cluster of the device identifiers having multiple overlapping characteristics of behavior to the confident computing devices from the first dataset, from the second dataset, wherein the device identifiers having the multiple overlapping characteristics of behavior are obtained when a plurality of first behavioral characteristics of the matched set overlap in comparison with a plurality of second behavioral characteristics of the second set.

15. The system of claim 10, wherein the processor further performs scoring a matched set of computing devices against a behavioral attribute by: generating a user scoring model based on a function of behavioral attributes of the matched set of computing devices; and assigning, using the user scoring model, a score for each of the matched set of computing devices against the behavioral attribute.

16. The system of claim 10, wherein the processor further performs: obtaining weights of a plurality of behavioral attributes from the client; configuring the trained deep learning model based on the weights to obtain a reconfigured deep learning model; and generating a cluster for a matched set of computing devices using the re-configured deep learning model.

17. The system of claim 10, wherein the trained machine learning model employs a classification method, the classification method depending on a level of learning between behavioral attributes of a matched set of computing devices and behavioral attributes of the second set of computing devices.

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